Pengcheng Liao

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Summary

Quantum information researcher with expertise in quantum error correction, variational and machine-learning based quantum algorithms, and quantum state/process tomography.

Technical Skills

- **Quantum information**: quantum error correction, stabilizer and graph states, bosonic encodings, non-Gaussianity, quantum tomography
- **Algorithms**: variational quantum algorithms, quantum machine learning, complexity analysis
- **Numerical & software**: Python, QuTiP, PyTorch; large-scale numerical simulation and data analysis

Education

University of Southern California Los Angeles, USA *Jan 2022–Present*
PhD in Engineering–Electrophysics (expected 2026) Advisor: Prof. Quntao Zhuang

University of Calgary Calgary, Canada *Sep 2019–Dec 2021*
MSc in Physics Advisors: Prof. David Feder & Prof. Barry Sanders

University of Chinese Academy of Sciences Beijing, China *Sep 2015–May 2019*
BSc in Physics

Publications

[1] Xiaobin Zhao, **Pengcheng Liao**, Francesco Anna Mele, Ulysse Chabaud, Quntao Zhuang. *Complexity of quantum tomography from genuine non-Gaussian entanglement.* Nature Communications (2025).

[2] **Pengcheng Liao**, Bingzhi Zhang, Quntao Zhuang. *Quantum-enhanced learning with a controllable bosonic variational sensor network.* Quantum Science and Technology 9, 045040 (2024).

[3] **Pengcheng Liao**, Quntao Zhuang. *Noisy entanglement testing for ranging and communication.* Physical Review Applied 22, 024007.

[4] **Pengcheng Liao**, Barry Sanders, Tim Byrnes. *Quadratic speedup for quantum perceptron training.* Physical Review A 110, 062412.

[5] **Pengcheng Liao**, Barry Sanders, David L. Feder. *Topological graph states and quantum error-correction codes.* Physical Review A 105, 042418.

[6] **Pengcheng Liao**, David L. Feder. *Graph-state representation of the toric code.* Physical Review A 104, 012432.

Conference

- 2025, Optica Sensing Congress — *Noisy Entanglement Testing for Ranging and Communication*
- 2024, APS March Meeting — *Quantum-enhanced physical-layer data learning with a variational sensor network*
- 2022, APS March Meeting — *Topological graph states and quantum error correction codes*
- 2021, The Fields Institute Workshop on Algebraic Graph Theory and Quantum Information —

Topological graph states

- 2021, APS March Meeting — *Quantum circuits for topological state generation*